

1. [A.] a) Describe the most popular wireless internet access technologies today. Compare and contrast them.

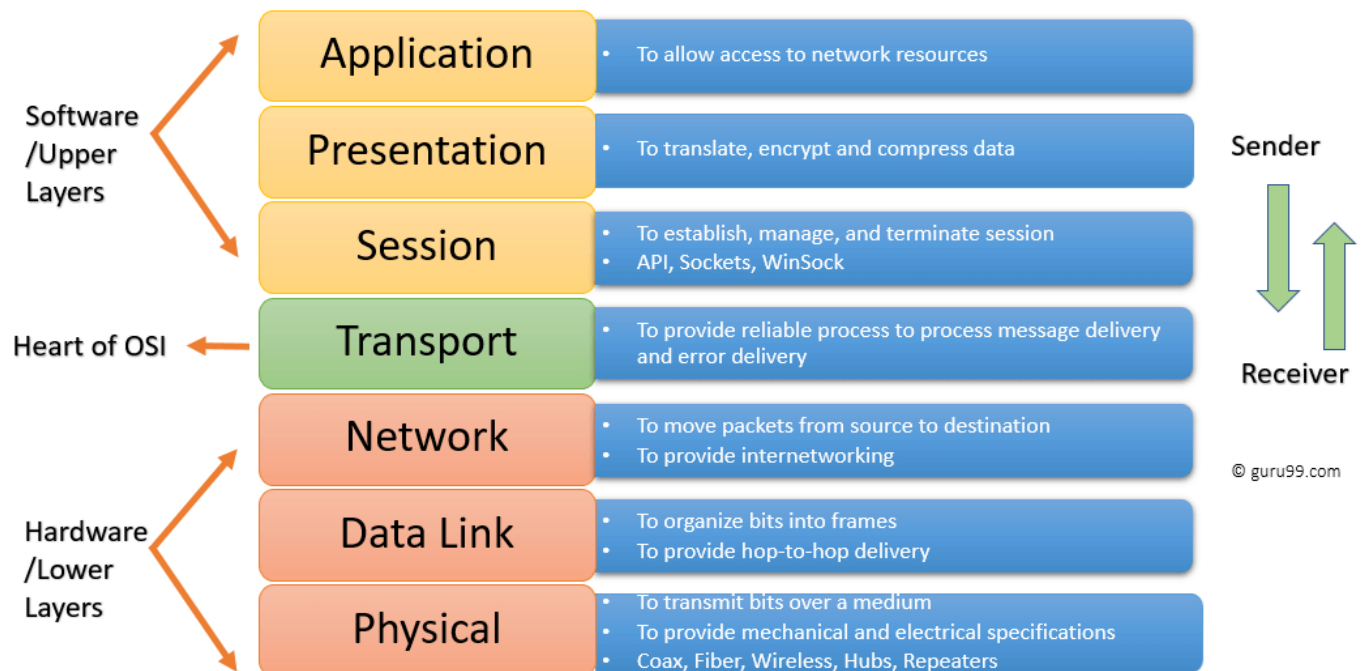
There are two popular wireless Internet access technologies today:

1. **Wifi (802.11)** In a wireless LAN, wireless users transmit/receive packets to/from a base station (i.e., wireless access point) within a radius of few tens of meters. The base station is typically connected to the wired Internet and thus serves to connect wireless users to the wired network.
2. **3G and 4G wide-area wireless access networks.** In these systems, packets are transmitted over the same wireless infrastructure used for cellular telephony, with the base station thus being managed by a telecommunications provider. This provides wireless access to users within a radius of tens of kilometers of the base station.

b) List four access technologies. Classify each one as home access, enterprise access, or wide-area wireless access.

1. Dial-up modem over telephone line: home;
2. DSL over telephone line: home or small office;
3. Cable to HFC: home;
4. 100 Mbps switched Ethernet: enterprise.

[B.] Write the duties/services and protocol of each specific OSI layer.



Layer	Name	Protocols
Layer 7	Application	SMTP, HTTP, FTP, POP3, SNMP
Layer 6	Presentation	MPEG, ASCH, SSL, TLS
Layer 5	Session	NetBIOS, SAP
Layer 4	Transport	TCP, UDP
Layer 3	Network	IPV5, IPV6, ICMP, IPSEC, ARP, MPLS.
Layer 2	Data Link	RAPA, PPP, Frame Relay, ATM, Fiber Cable, etc.
Layer 1	Physical	RS232, 100BaseTX, ISDN, 11

[C.] Consider an application that transmits data at a steady rate (for example, the sender generates an N-bit unit of data every k time unites, where k is small and fixed). Also, when such an application starts, it will continue running for a relatively long period of time. Answer the following questions, briefly justifying your answer:

a. Would a packet-switched network or a circuit-switched network be more appropriate for this application? why?

A circuit-switched network would be well suited to the application, because the application involves long sessions with predictable smooth bandwidth requirements. Since the transmission rate is known and not bursty, bandwidth can be reserved for each application session without significant waste. In addition, the overhead costs of setting up and tearing down connections are amortized over the lengthy duration of a typical application session.

b. Suppose that a packet-switched network is used and the only traffic in this network comes from such applications as described above. Furthermore, assume that the sum of the application data rates is less than the capacities of each and every link. Is some form of congestion control needed? Why?

In the worst case, all the applications simultaneously transmit over one or more network links. However, since each link has sufficient bandwidth to handle the sum of all of the applications' data rates, no congestion (very little queuing) will occur. Given such generous link capacities, the network does not need congestion control mechanisms.

[D.] Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, or rates $R_1 = 500$ kbps, $R_2 = 2$ Mbps and $R_3 = 1$ Mbps.

a. Assuming no other traffic in the network, what is the throughput for the file transfer?

500kbps

b. Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will take to transfer the file to Host B?

64 seconds

c. Repeat (a) and (b) but now with R2 reduced to 100 kbps.

100kbps, 320 second

2.[A.] How are the blocks allocated? What are the restrictions need to be applied to the allocated block.

Block Allocation

The next issue in classless addressing is block allocation. How are the blocks allocated? The ultimate responsibility of block allocation is given to a global authority called the Internet Corporation for Assigned Names and Addresses (ICANN). However, ICANN does not normally allocate addresses to individual Internet users. It assigns a large block of addresses to an ISP (or a larger organization that is considered an ISP in this case). For the proper operation of the CIDR, three restrictions need to be applied to the allocated block.

1. The number of requested addresses, N , needs to be a power of 2. This is needed to provide an integer value for the prefix length, n (see the second restriction). The number of addresses can be 1, 2, 4, 8, 16, and so on.
2. The value of prefix length can be found from the number of addresses in the block. Since $N = 2^{32-n}$, then $n = \log_2 (2^{32}/N) = 32 - \log_2 N$. That is the reason why N needs to be a power of 2.
3. The requested block needs to be allocated where there are a contiguous number of unallocated addresses in the address space. However, there is a restriction on choosing the beginning addresses of the block. The beginning address needs to be divisible by the number of addresses in the block. To see this restriction, we can show that the beginning address can be calculated as $X \times 2^{n-32}$ in which X is the decimal value of the prefix. In other words, the beginning address is $X \times N$.

[B.] One of the addresses in a block is 110.23.120.14/20. Find the number of addresses, the first address and the last address in the block.

The network mask is 255.255.240.0.

The number of addresses in the network is $2^{32-20} = 4096$.

- To find the first address, we apply the first short cut to bytes 1, 2, and 4 and the second short cut to byte 3. The first address is 110.23.112.0/20.

Address:	110	.	23	.	120	.	14
Network mask:	255	.	255	.	240	.	0
First address (AND):	110	.	23	.	112	.	0

- To find the last address, we apply the first short cut to bytes 1, 2, and 4 and the second short cut to byte 3. The OR operation is applied to the complement of the mask. The last address is 110.23.127.255/20.

Address:	110	.	23	.	120	.	14
Network mask:	0	.	0	.	15	.	255
Last address (OR):	110	.	23	.	127	.	255

[C.] An organization is granted a block of addresses with the beginning address 254.24.252.0/24. The organization needs to have 4 sub-blocks of addresses to use in its four subnets as shown below:

- ✓ One subblock of 123 addresses
- ✓ One subblock of 62 addresses
- ✓ One subblock of 11 addresses
- ✓ One subblock of 5 addresses

Design the sub-blocks.

The answer is like this, modify it as question requirement.

Solution: Allocated no. of address: $2^{32-24} = 256$

First address: 14.24.74.0/24;

Last address: 14.24.74.255/24

Mask: 255.255.255.0

We should start with largest sub-blocks.

❖ $N_1=120 \Rightarrow N_1=128 \Rightarrow n_1=32-\log_2 128 = 25$

First address: 14.24.74.0/25

Last address: 14.24.74.127/25 Mask: 255.255.255.128 (as last octet: 1000 0000)

❖ $N_2=60 \Rightarrow N_2=64 \Rightarrow n_2=32-\log_2 64 = 26$

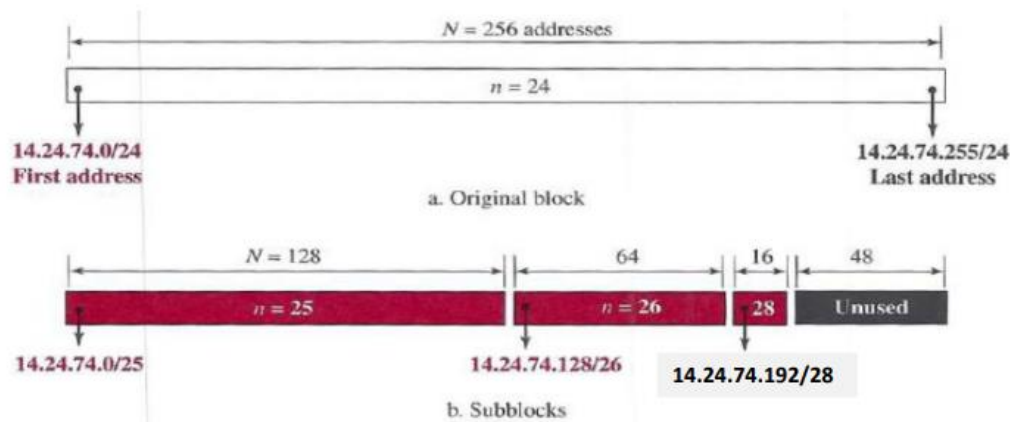
First address: 14.24.74.128/26

Last address: 14.24.74.191/26 Mask: 255.255.255.192 (as last octet: 1100 0000)

❖ $N_3=10 \Rightarrow N_3=16 \Rightarrow n_3=32-\log_2 16 = 28$

First address: 14.24.74.192/28

Last address: 14.24.74.207/28 Mask: 255.255.255.240 (as last octet: 1111 0000)



[D.] Write the steps need to be carefully followed to guarantee the proper operation of the subnetworks.

The sub networks in a network should be carefully designed to enable the routing of packets. We assume the total number of addresses granted to the organization is N , the prefix length is n , the assigned number of addresses to each subnetwork is N_{sub} , the prefix length for each subnetwork is n_{sub} , and the total number of subnetworks is s .

Then, the following steps need to be carefully followed to guarantee the proper operation of the subnetworks.

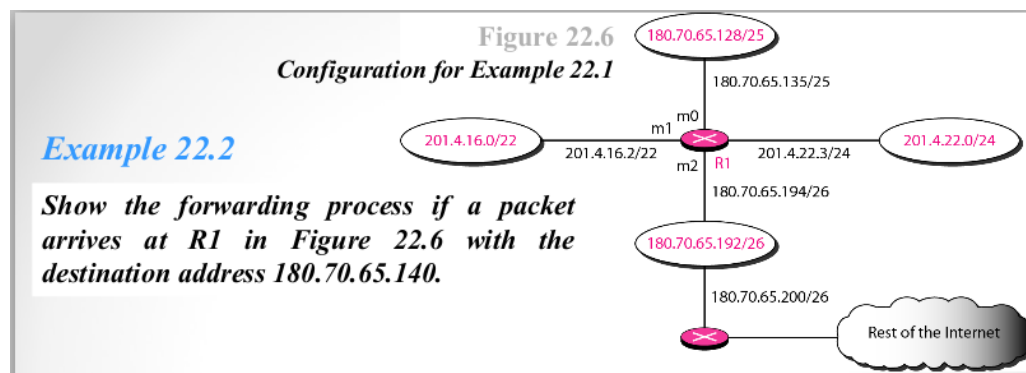
1. The number of addresses in each subnetwork should be a power of 2.
2. The prefix length for each subnetwork should be found using the following formula:
$$n_{\text{sub}} = n + \log_2(N/N_{\text{sub}})$$
3. The starting address in each subnetwork should be divisible by the number of addresses in that subnetwork. This can be achieved if we first assign addresses to larger networks. The restrictions applied in allocating addresses for a subnetwork are parallel to the ones used to allocate addresses for a network.

3.[A.] Discuss the packet forwarding without subnetting.

The following steps describe the forwarding module that is used to forward packets in the classful addressing where subnetting is not involved.

1. From the packets header, the destination address is determined.
2. The destination address is used to determine the class of the address.
3. After identifying the class of the address, the destination address and class are combinedly used to extract the network address.
4. Now the class of the destination address is used to decide which table among the three must be referred and the network address extracted is used to determine the next-hop and interface number.
5. The next-hop (logical address of next router) and interface number are provided to ARP to determine the physical address of the next router.

[B.] Show the forwarding process if a packet arrives at R1 in below figure with the destination address 180.70.65.140.



The router performs the following steps:

1. The first mask (/26) is applied to the destination address. The result is 180.70.65.128, which does not match the corresponding network address.
2. The second mask (/25) is applied to the destination address. The result is 180.70.65.128, which matches the corresponding network address. The next-hop address and the interface number m0 are passed to ARP for further processing.

[C.] Compare and contrast the properties of a centralized and distributed routing algorithm. Give an example of routing that takes a centralized and a decentralized approach.

A centralized routing algorithm computes the least-cost path between a source and destination by using complete, global knowledge about the network. The algorithm needs to have the complete knowledge of the connectivity between all nodes and all links' costs. The actual calculation can be run at one site or could be replicated in the routing component of each and every router. A distributed routing algorithm calculates the least-cost path in an iterative, distributed manner by the routers. With a decentralized algorithm, no node has the complete information about the costs of all network links. Each node begins with only the knowledge of the costs of its own directly attached links, and then through an iterative process of calculation and information exchange with its neighboring nodes, a node gradually calculates the least-cost path to a destination or a set of destinations. OSPF protocol is an example of centralized routing algorithm, and BGP is an example of a distributed routing algorithm.

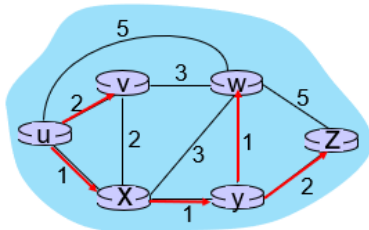
[D.] Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from u to all network nodes. Show how the algorithm works by computing a table.

Algorithm:

```
1  Initialization:
2   $N' = \{u\}$                                 /* compute least cost path from u to all other nodes */
3  for all nodes v
4    if v adjacent to u                        /* u initially knows direct-path-cost only to direct neighbors */
5      then  $D(v) = c_{u,v}$                       /* but may not be minimum cost! */
6    else  $D(v) = \infty$ 
7
8  Loop
9    find w not in  $N'$  such that  $D(w)$  is a minimum
10   add w to  $N'$ 
11   update  $D(v)$  for all v adjacent to w and not in  $N'$  :
12      $D(v) = \min ( D(v), D(w) + c_{w,v} )$ 
13   /* new least-path-cost to v is either old least-cost-path to v or known
14      least-cost-path to w plus direct-cost from w to v */
15  until all nodes in  $N'$ 
```



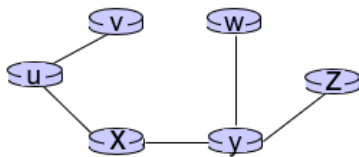
Step	N'	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x	2,x	∞	∞
2	uxy	2,u	3,y	3,y	4,y	∞
3	uxyv	2,u	3,y	3,y	4,y	∞
4	uxyvw	2,u	3,y	3,y	4,y	4,y
5	uxyvwz	2,u	3,y	3,y	4,y	4,y



Initialization (step 0): For all a : if a adjacent to u then $D(a) = c_{u,a}$

find a not in N' such that $D(a)$ is a minimum
add a to N'
update $D(b)$ for all b adjacent to a and not in N' :
 $D(b) = \min (D(b), D(a) + c_{a,b})$

resulting least-cost-path tree from u :



resulting forwarding table in u :

destination	outgoing link
v	(u,v)
x	(u,x)
y	(u,x)
w	(u,x)
z	(u,x)

route from u to v directly

route from u to all other destinations via x

1.

A. The following is a dump of a UDP header in hexadecimal format.

0045DF000058FE20

a) What is the source port number?

The source port number is the first 2 bytes (4 hexadecimal digits). In this case, it is 0045, which is 69 in decimal.

b) What is the destination port number?

The destination port number is the next 2 bytes (4 hexadecimal digits). In this case, it is DF00, which is 57188 in decimal.

c) What is the total length of the user datagram?

The total length of the user datagram is the next 2 bytes (4 hexadecimal digits). In this case, it is 0058, which is 88 in decimal.

d) What is the length of the data?

The length of the data can be calculated by subtracting the UDP header length (8 bytes) from the total length. So, 88 (total length) - 8 (UDP header length) = 80 bytes.

e) Is the packet directed from a client to a server or vice versa?

To determine the direction of the packet (client to server or vice versa), you typically need additional information from the network capture or protocol

specification. The UDP header itself does not provide this information. You would need to know the context in which this packet was captured.

f) What is the client process?

The UDP header does not provide information about the specific client process. Information about the client process would typically be found in the application layer protocols (e.g., in the payload data), not in the UDP header.

B. How do opening, closing and shrinking of window occur in TCP? Give examples for both send receive window.

Opening of Windows:

- **Sender Window (Send Window):** Starts small and increases as acknowledgments are received.
 - ✓ **Example:** Initially set to 10 segments, may increase to 15 as acknowledgments arrive.
- **Receiver Window (Receive Window):** Represents buffer space; increases as data is processed.
 - ✓ **Example:** Initially set to 20 segments, may increase to 25 as data is acknowledged.

Closing of Windows:

- **Sender Window (Send Window):** May close or shrink due to congestion or network issues.
 - ✓ **Example:** Reduces from 20 to 15 during congestion.
- **Receiver Window (Receive Window):** Closes when the receiver's buffer is full.
 - ✓ **Example:** Closes when it has acknowledged 30 packets and needs to process them.

C. what are the services that a transport-layer protocol can offer to applications invoking it? Describe each with example.

- **End-to-End Delivery:** Ensuring data is reliably delivered from the source to the destination.
 - ✓ **Example:** TCP guarantees end-to-end delivery of web pages, emails, and files by retransmitting lost packets.
- **Addressing:** Identifying the source and destination of data using addresses or ports.
 - ✓ **Example:** In TCP/IP, source and destination IP addresses and port numbers are used to address data packets.
- **Reliable Delivery:** Ensuring data arrives without errors and in the correct order.
 - ✓ **Example:** TCP provides reliable delivery by using acknowledgments and retransmissions for error-free file transfers.
- **Flow Control:** Managing the rate of data transmission to prevent congestion and ensure efficient delivery.
 - ✓ **Example:** TCP employs flow control mechanisms like windowing to regulate data flow between sender and receiver.
- **Multiplexing:** Supporting multiple applications on the same network connection.

- ✓ **Example:** TCP and UDP use port numbers to multiplex data, allowing multiple applications like web browsers and email clients to use a single network link simultaneously.

D. A sender sends a JPEG message. Show the MIME header.

The MIME (Multipurpose Internet Mail Extensions) header for a JPEG message typically looks like this:

Content-Type: image/jpeg

This header indicates that the content of the message is in JPEG format.

2.

- A. Suppose, An FTP server has received a packet from an FTP client with IP address 153.2.7.9. The FTP server wants to verify that the FTP client is an authorized client. The FTP server can consult a file containing the list of authorized clients. However, the file consists only of domain names. The FTP server has only the IP address of the requesting client, which was the source IP address in the received IP datagram. The FTP server asks the resolver (DNS client) to send an inverse query to a DNS server to ask for the name of the FTP client. Show the query and response messages with values for each field. Discuss separately.**

0x1200		0x0900	
1		0	
0		0	
1	'9'	1	'7'
1	'2'	3	'1'
'5'	'3'	7	'i'
'n'	'-'	'a'	'd'
'd'	'r'	4	'a'
'r'	'p'	'a'	0
12		1	

The first 2 bytes show the identifier (0x1200). The flags value is 0x0900 in hexadecimal. In binary it is 0000100100000000, and we divide it into fields as shown below:

QR	OpCode	AA	TC	RD	RA	Reserved	rCode
0	0001	0	0	1	0	000	0000

The OpCode is 0001, which defines an inverse query. The message contains only one question record. The domain name is 19171231537in-addr4arpa. The next 2 bytes define the query type as PTR, and the last 2 bytes define the class as the Internet.

Figure 19.24 shows the response. The flags value is 0x8D80 in hexadecimal. In binary it is 1000110110000000, and we divide it into fields as shown below:

QR	OpCode	AA	TC	RD	RA	Reserved	rCode
1	0001	1	0	1	0	000	0000

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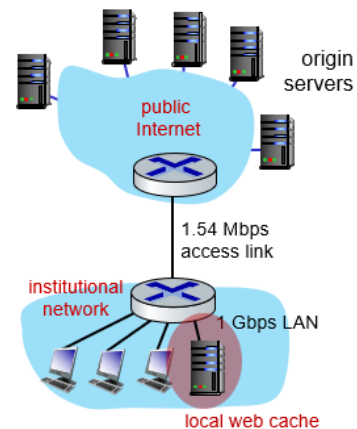
B. Describe how web caching can reduce the delay in receiving a requested object. Will web caching reduce the delay for all objects requested by a user or for only some of the objects? Why?

Calculating access link utilization, end-end delay with cache:

suppose cache hit rate is 0.4:

- 40% requests served by cache, with low (msec) delay
- 60% requests satisfied at origin
 - rate to browsers over access link
 $= 0.6 * 1.50 \text{ Mbps} = .9 \text{ Mbps}$
 - access link utilization $= 0.9 / 1.54 = .58$ means low (msec) queueing delay at access link
- average end-end delay:
 - $= 0.6 * (\text{delay from origin servers})$
 - $+ 0.4 * (\text{delay when satisfied at cache})$
 - $= 0.6 (2.01) + 0.4 (\sim \text{msecs}) = \sim 1.2 \text{ secs}$

lower average end-end delay than with 154 Mbps link (and cheaper too!)



Web caching can reduce the delay for all objects, even objects that are not cached, since caching reduces the traffic on links.

C. What are the services provided by the SSH protocol through port forwarding? Explain with example.

- **Local Port Forwarding:** Redirects traffic from a local port on the client to a remote server.
 - ✓ **Example:** Forwarding local port 8080 to a remote web server's port 80. This allows accessing the remote web server through **localhost:8080** on the client.
- **Remote Port Forwarding:** Redirects traffic from a remote port on the server to a local client.
 - ✓ **Example:** Forwarding remote port 3306 on the server to the client's port 3306. This allows the server to connect to a database running on the client.
- **Dynamic Port Forwarding (SOCKS Proxy):** Creates a dynamic SOCKS proxy on the client, allowing multiple connections to be forwarded through a single SSH tunnel.
 - ✓ **Example:** Establishing a dynamic port forwarding tunnel and configuring a web browser to use it as a SOCKS proxy. This enables secure and anonymous web browsing through the SSH tunnel.

D. Show the encoding for the INTEGER 1456 in case of SNMP. (Chapter-28)

3.

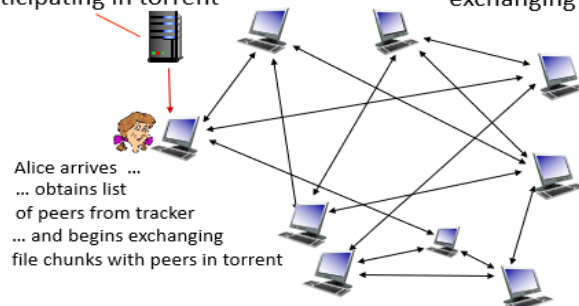
A. Suppose a content provider, NetCinema, employs the third-party CDN company, KingCDN, to distribute its videos to its customers. On the NetCinema web pages, each of its videos is assigned a URL that includes the string "video" and a unique identifier for the video itself; for example, Transformer 7 might be assigned <http://video.netcinema.com/6Y7B23V> . If DASH is used , what will be the steps before the client can dynamically select chunks from the different versions of the video?(Chapter-25).

B. Consider a new peer Alice that joins BitTorrent without possessing any chunks without any chunks, she cannot become a top-four uploader for any of the other peers, since she has nothing to upload. How then will Alice get her first chunk?

- file divided into 256Kb chunks
- peers in torrent send/receive file chunks

tracker: tracks peers participating in torrent

torrent: group of peers exchanging chunks of a file



Recall that in BitTorrent, a peer picks a random peer and 'optimistically unchokes' the peer for a short period of time. Therefore, Alice will eventually be optimistically unchoked by one of her neighbors, during which time she will receive chunks from that neighbor.

C. Define Jitter. Explain the application to TFTP in conjunction with DHCP.

Jitter is the variation in the delay of received packets in a network, particularly in packet-switched networks like the Internet.

In conjunction with DHCP (Dynamic Host Configuration Protocol), TFTP (Trivial File Transfer Protocol) is often used during the boot process of networked devices, such as computers and network appliances, to retrieve configuration or boot files. Here's a brief explanation of their application together:

- **DHCP Configuration:** DHCP dynamically assigns IP addresses and network configuration settings to devices when they connect to a network. This includes information like the device's IP address, subnet mask, default gateway, and DNS server addresses.
- **TFTP Boot:** In some cases, DHCP can also provide information about a TFTP server and a specific file to retrieve. This is particularly useful in scenarios where devices need to boot from a network, such as thin clients, network-attached storage (NAS) devices, or diskless workstations.
- **TFTP File Retrieval:** The device, upon receiving the DHCP-assigned information, knows that it should retrieve a specific file from the TFTP server mentioned in the DHCP response. It uses TFTP to request and download this file from the server.
- **File Examples:** The file retrieved via TFTP could be various things, such as a boot loader, an operating system kernel, or configuration files needed for the device to function properly within the network.

D. Do you think H.323 is actually the same as SIP? Make a comparison between the two.

No, H.323 and SIP (Session Initiation Protocol) are not the same; they are two different protocols used for real-time communication over the Internet and IP-based networks.

Here's a comparison between the two:

1. H.323 protocol is based on telephony while SIP is based on internet.
2. The H.323 standard body is ITU. In contrast, SIP is designed by IETF.
3. H.323 follows binary message format. Conversely, SIP uses ASCII message format.
4. H.323 is not compatible with internet while SIP has the compatibility with the internet.
5. The architecture of H.323 is monolithic. Inversely, SIP is built on the modular architecture.
6. SIP offers the instant messaging facility. On the contrary, there is no such facility in H.323.